

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

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COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I Preethi R/192511352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Preethi

$25 + 25 + 19 + 19 + 9$
 $= 97$

Student 25: Preethi R | Register Number: 192511352

Question 1

Case Study - Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Question 2

Case Study - Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

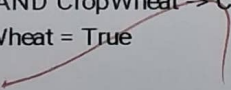
- I. for-all x : Vehicle(x) AND EmergencyType(x) $\rightarrow$ ClearPath(x) [Rule 1]
- II. for-all x : ClearPath(x) $\rightarrow$ GreenSignal(x) AND NOT RedSignal(x) [Rule 2]
- III. for-all x for-all y : Vehicle(x) AND Behind(x, y) AND GreenSignal(y) $\rightarrow$ Proceed(x) [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed(CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Question 3

Case Study - Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
 - 2) P2: IrrigationNeeded AND CropWheat $\rightarrow$ ApplyDripMethod
 - 3) P3: NOT ApplyDripMethod AND CropWheat $\rightarrow$ CropAtRisk
 - 4) Facts: SoilDry = True, CropWheat = True
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- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move NOT inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Question 4

Case Study - Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench[1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze[1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter[x,y] $\rightarrow$ Gold is at [x,y]
- 4) NOT Wumpus[x,y] AND NOT Pit[x,y] $\rightarrow$ Safe[x,y]

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.
- (c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

Smart Medical Diagnosis System:

a) Propositional Logic Representation:

Let:

F = patient A has Fever

C = Patient A has Cough

R = patient A has Rash

B = patient A has Breathlessness

Flu = patient A has possible Flu

M = patient A has possible Measles

P = patient A has possible pneumonia

Rules:

1. Fever AND Cough $\rightarrow$ possible Flu

$$F \wedge C \rightarrow \text{Flu}$$

2. Fever AND Rash $\rightarrow$ possible Measles

$$F \wedge R \rightarrow M$$

3. Cough AND Breathlessness $\rightarrow$ possible pneumonia

$$C \wedge B \rightarrow P$$

Facts:

F = True

C = True

B = True.

b) Forward chaining

Forward chaining starts with the known facts and applies rules whose condition are satisfied.

Initial facts:

Fever, Cough, Breathlessness.

Step 1:

rule 1: $F \wedge C \rightarrow Flu$

We know:

$F = True$

$C = True$

Therefore:

$Flu = True$.

Step 2:

$F \wedge R \rightarrow M$

We know Fever is true but Rash is not given.

Therefore, rule 2 cannot fire.

Step 3:

$C \wedge B \rightarrow P$

We know: $C = True$ and $B = True$.

Therefore:

Pneumonia = True.

Final derived facts

1. $F \wedge C \rightarrow Flu$ possible flu
2. $F \wedge R \rightarrow M$ cannot fire! Rash unavailable.
3. $C \wedge B \rightarrow P$ possible pneumonia

patient A has possible flu and possible pneumonia.

c) Backward Chaining:

Goal

$P =$ patient A has pneumonia

Goal: P

Find a rule whose conclusion is P :

$C \wedge B \rightarrow P$

Therefore, reduce the goal into two subgoals.

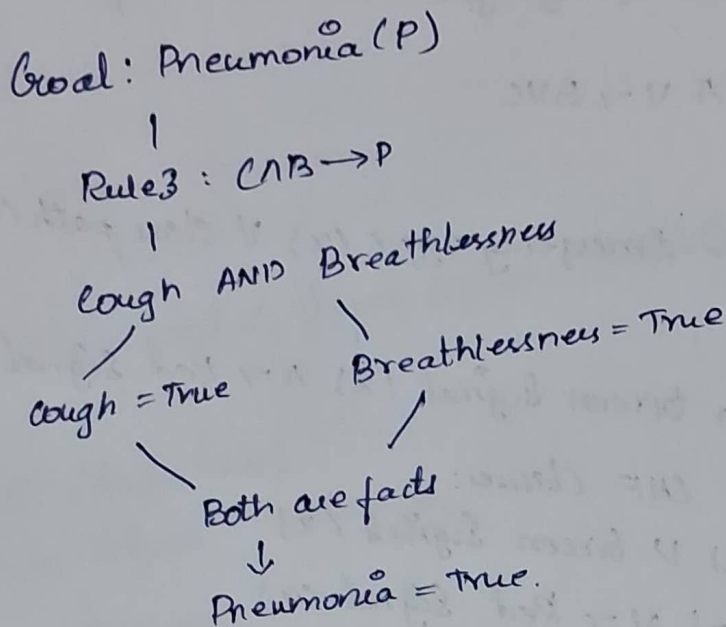
$C \wedge B \rightarrow P$.

CNB = True

Hence:

P = True.

Goal reduction Tree



2. Autonomous Traffic Management Agent

a) Unification for rule 1

$\forall x \text{ Vehicle}(x) \wedge \text{Emergency Type}(x) \rightarrow \text{Clear path}(x)$

Given facts:

→ Vehicles (Ambulance)

→ Emergency Type (Ambulance)

First predicate: Vehicle(x) with Vehicle (Ambulance)

Therefore: x / Ambulance

Second predicate:

Emergency Type(x) becomes Emergency Type (Ambulance)

This is also a fact.

MGU

$\theta = \{x / \text{Ambulance}\}$

clear path (Ambulance) is desired.

b) Resolution proof of proceed [Car A]

Convert rules to CNF

$\text{Vehicle}(x) \wedge \text{Emergency Type}(x) \rightarrow \text{clear path}(x)$

using:

$$A \wedge B \rightarrow C \equiv \neg A \vee \neg B \vee C$$

Therefore:

$\neg \text{Vehicle}(x) \vee \neg \text{Emergency Type}(x) \vee \text{clear path}(x)$

rule 2:

$\text{clear path}(x) \rightarrow \text{Green Signal}(x) \wedge \neg \text{Red Signal}(x)$

This produces two CNF clauses:

$\neg \text{clear path}(x) \vee \text{Green Signal}(x)$

$\neg \text{clear path}(x) \vee \neg \text{Red Signal}(x)$

rule 3:

$\text{Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{Green Signal}(y) \rightarrow \text{path}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{Green Signal}(y)$

Facts:

Vehicle (Ambulance)

Emergency Type (Ambulance)

Vehicle (Car A)

Behind (Car A, Ambulance)

Derive clear path (Ambulance)

from:

$\neg \text{Vehicle}(x) \vee \neg \text{Emergency Type}(x) \vee \text{clear path}(x)$

and

Vehicle (Ambulance)

Resolution:

→ Emergency Type (Ambulance) $\vee$ clear path (Ambulance)

Resolve:

Emergency Type (Ambulance)

Therefore:

Clear path (Ambulance)

Derive Green Signal (Ambulance)

Use rule 2:

→ Clear path (Ambulance) $\vee$ Green Signal (Ambulance)

Resolve with:

Clear path (Ambulance)

Therefore: Green Signal (Ambulance)

Derive proceed (car A)

rule 3:

→ Vehicle (car A) $\vee$ → Behind (car A, Ambulance) $\vee$ Green Signal

Resolution: proceed (car A)

Resolution chain

Vehicle (Ambulance) + Emergency Type (Ambulance)

↓ rule 1

~~Clear path (Ambulance)~~

↓ rule 2

Green Signal (Ambulance)

+

Vehicle (car A) + Behind (car A, Ambulance)

↓ rule 3

proceed (car A)

3. Agricultural Expert Reasoning System

a) CNF Conversion

P_1 :

Soil Dry $\rightarrow$ Irrigation needed

Eliminate implication:

$\rightarrow$ Soil Dry $\vee$ Irrigation needed

P_2 :

Irrigation needed $\wedge$ Crop wheat $\rightarrow$ Apply Drip method.

$\rightarrow$ Irrigation needed $\vee \rightarrow$ Crop wheat $\vee$ Apply Drip method.

P_3 :

$\rightarrow$ Apply drip method $\wedge$ Crop wheat $\rightarrow$ Crop At risk

Apply drip method $\vee \rightarrow$ Crop wheat $\vee$ Crop at risk

Facts:

Soil dry, Crop wheat

b) Resolution refutation

Goal: Apply Drip method

Negative goal: $\neg$ Apply Drip method

Resolution Steps:

1. Soil Dry + $\neg$ Soil dry $\vee$ Irrigation needed

$\Rightarrow$ Irrigation needed

2. Irrigation needed + $\neg$ Irrigation needed $\vee \rightarrow$ Crop wheat $\vee$ Apply drip method

$\Rightarrow \rightarrow$ Crop wheat $\vee$ Apply drip method

3. Crop wheat + above clause:

$\Rightarrow$ Apply drip method.

3) Remove Soiddy

without Soiddy:

- Irrigation needed cannot be derived
- Apply drip method cannot be derived
- Apply drip method is also not automatically true.
- Crop at risk cannot be concluded.

Crop at risk does not follow

4) Klumpus World.

a) Belief State

- Stench [1,2]
- Breeze [1,1]
- Glitter [2,2]

Stench [1,2] → Klumpus adjacent to [1,2]

Breeze [1,1] → pit adjacent to [1,1]

Glitter [2,2] → Gold [2,2]

Therefore: Gold [2,2] and possible dangers are adjacent to the corresponding cells.

b) Forward chaining

From Stench [1,2]:

possible Klumpus location:

[1,1], [1,3], [2,2]

From Breeze $[1,1]$:

possible pit location:

$[1,2], [2,1]$

From Clutter $[2,2]$: Gold $[2,2]$

c) path safety:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

→ $[1,2]$ may contain a pit

→ $[2,2]$ may contain the Mumpus

→ The KB does not contain enough information to prove these cells are safe.

The path cannot be confirmed as safe.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

25 + 25 + 20 + 20 + 10 = 100

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